FPDDet: An Efficient Rotated SAR Ship Detector Based on Simple Polar Encoding and Decoding

Moran Ju, Buniu Niu, and Jingbo Zhang

***Abstract*— In the task of synthetic aperture radar (SAR) ship detection, ship targets exhibit arbitrary orientations and are closely aligned. In recent years, oriented bounding box (OBB)-based detectors have gained attention as a solution to the significant overlap issue present in horizontal bounding box (HBB)-based detectors. Due to the boundary discontinuity issue with current mainstream OBB methods, detectors utilizing the polar coordinate system to define OBB have been introduced. However, currently available methods are usually complicated to encode and decode and do not consider the complex background of SAR images. This results in the complexity of neural networks and inaccurate predictions. In this article, we propose a novel five-parameter polar coordinate dense regression detector (FPDDet) that only uses a centroid, a mapped polar diameter, and two polar angles to deal with the overlap problem of HBB and the boundary discontinuity problem of OBB. Meanwhile, we introduce a dense regression strategy based on covariance-adaptive rotated Gaussian heatmap to dynamically assign ship target samples in response to large-scale variation of ship targets in SAR images, and we suggest a dense regression heatmap loss function to better match our dense regression strategy. In addition, we design a feature enhancement (FE) module to enhance the target features while weakening the background interference, aiming to cope with the severe noise pollution of SAR images. Experimental results show that our FPDDet achieves the state-of-the-art performance on rotating SAR ship detection dataset (RSSDD) and rotated ship detection dataset (RSDD). Compared to previous best results, mAP has been improved by 1.2% and 1.71%.**

***Index Terms*— Arbitrary oriented, polar encoding, ship detection, single stage, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) image possesses all-weather and all-day imaging capabilities and remains unaffected by cloud cover, rendering it an indispensable ground observation method for applications such as maritime monitoring [1], [2], ship discrimination [3], and flood detection [4]. The present obstacles faced by SAR image ship detection can be concluded as follows.

Manuscript received 20 June 2023; revised 23 August 2023; accepted 9 October 2023. Date of publication 13 October 2023; date of current version 27 October 2023. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant 3132023233 and in part by the National Natural Science Foundation of China under Grant 62201114 and Grant 62071080. *(Corresponding author: Buniu Niu.)*

The authors are with the College of Information Science and Technology, Dalian Maritime University, Dalian 116026, China (e-mail: jumoran@dlmu.edu.cn; niubuniu@dlmu.edu.cn; zhang_jingbo@dlmu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3324355

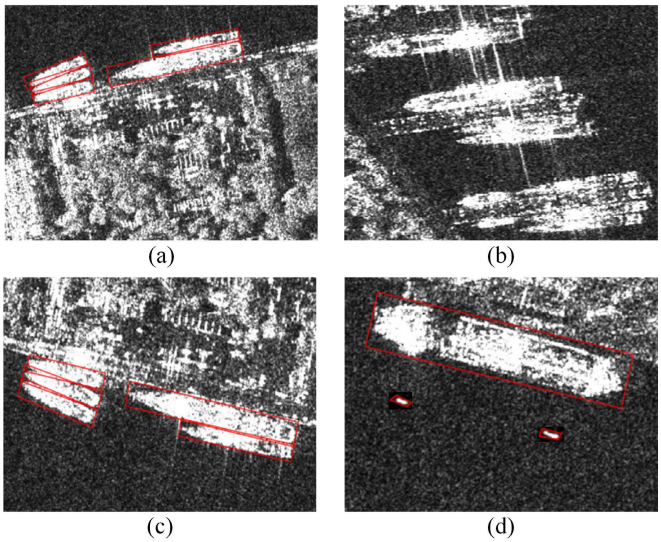


Fig. 1. Obstacles faced by SAR image object detection. (a) Complex image background. (b) Blurred ship boundaries. (c) Dense arrangement of ships. (d) Large-scale variations.

1) *Complex Image Backgrounds, as Shown in* Fig. 1(a)*:* such as the various buildings on the shore and the inherent speckle noise in the SAR image, etc.
2) *Blurred Ship Target Boundaries, as Shown in* Fig. 1(b)*:* image blur caused by phase errors during imaging is a challenge for SAR object detection.
3) *Dense Inshore Ship Arrangements, as Shown in* Fig. 1(c)*:* a large number of ships alongside the coast make it difficult for the detectors to discriminate correctly.
4) *Large-Scale Variations of Ship Targets, as Shown in* Fig. 1(d)*:* for instance, the vast difference in scale between cruise ships and fishing boats poses a challenge to the network's feature extraction capability.

The abovementioned problems present challenges to traditional SAR ship detection methods based on handcrafted features [5], statistical distributions [6], and thresholds [7].

Based on the above problems faced by traditional SAR ship detection methods, unsupervised [8], [9], semisupervised [10], [11], [12], [13], [14], and supervised [15], [16], [17] learnable SAR ship detection methods can effectively alleviate the problem of traditional methods that rely heavily on a priori knowledge and improve the accuracy of ship detection. Among them, unsupervised methods can predict SAR ship objects

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

without labels, but the accuracy is lower compared to supervised and semisupervised methods, semisupervised methods rely on a small number of labels to significantly improve the prediction performance of SAR ship objects compared to unsupervised methods, and supervised methods have the best prediction performance compared to the other two types of methods but rely heavily on a large number of ship labels. With the successful application of convolutional neural network (CNN) in object detection and expanding datasets of SAR ships, CNN in SAR image ship object detection has also been rapidly developed recently. Current methods [15], [16], [17] based on horizontal bounding box (HBB) show good performance in detecting offshore ship targets, but the performance is not satisfactory for predicting inshore SAR images with complex backgrounds and densely arranged ship targets with large-scale variations. The difficulty in screening positive samples due to the large overlap of detection results makes it difficult to improve the localization accuracy of the HBB-based detector.

Recently, with the emergence of OBB-based detectors, the issue of accuracy degradation caused by overlapping prediction of densely arranged HBB has been partially resolved. However, these OBB-based methods introduce new problems, for example, the OBB-based detectors [18], [19], [20], [21] using centroid coordinates, width, height, and rotation angle cause the problem of boundary discontinuity, and the OBB represented by an arbitrary quadrilateral composed of four corner points, such as [22], [23], and [24], causes the problem of corner point disorder, which results in even a small deviation between the predicted box and the ground truth box causing an abrupt change in the loss and thus leading to instability during CNN training. To reduce the impact of the boundary discontinuity of the OBB-based detectors, Zhao et al. [25] and He et al. [26] introduced the polar coordinate system to represent the rotation target, where PolarDet [25] uses a centroid, a short edge, four corner points corresponding to the polar angle, and four polar diameters from the centroid to the corner points for the OBB regression, He et al. [26] proposed to use a set of vectors from the centroid to the OBB boundary to represent the rotation target. However, the above polar coordinate-based detectors increase the complexity of the model due to the regression of too many parameters and reduce the inference speed of the CNN because of the complexity of the encoding and decoding process. Moreover, these methods fail to account for the impact of factors, such as complex image background and the decline in prediction accuracy resulting from large-scale variations of ship targets.

In this article, we propose a novel five-parameter polar coordinate dense regression detector (FPDDet) for oriented SAR ship detection. Specifically, as shown in Fig. 2(c), we use a centroid of OBB, a polar diameter from the centroid to the corresponding minimum enclosing rectangular corner of OBB, and two polar angles to facilitate and efficiently encode the ship target in SAR images. For the decoding process of five-parameter polar coordinates that is similarly facilitated, as shown in Fig. 2(e), first, the predicted center of the target is obtained; then, the two target boundary points can be obtained from the predicted polar diameter and polar angles; and finally,

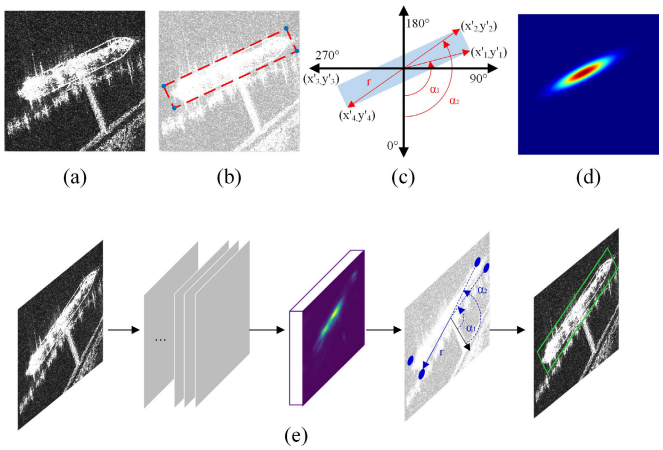


Fig. 2. (a) SAR image ship in arbitrary direction. (b) Label of the ship target. (c) Five-parameter polar coordinate encoding. (d) Covariance-adaptive rotated Gaussian heatmap of the ship target. (e) Decoding process for FPDDet.

the target prediction box can be quickly obtained according to the symmetry. Furthermore, due to the large variation of target scales in SAR images, we introduce an SDR strategy based on the covariance-adaptive rotated Gaussian heatmap as shown in Fig. 2(d) to dynamically assign the positive samples to ship targets of different scales. Meanwhile, to discriminate between densely arranged ship targets inshore, a dense regression heatmap loss function is proposed to learn the global confidence map of ship targets. Furthermore, based on the fact that inshore SAR images contain complex backgrounds and extremely blurred boundaries between ship targets and backgrounds, we put forward the SAR image target enhancement module to reduce the impact of the background on the prediction results and enhance the features of the ship targets. To verify the detection performance of FPDDet, we conduct the comparative experiments with classical oriented object detectors on rotating SAR ship detection dataset (RSSDD). The experimental results demonstrate the superior performance and efficiency of FPDDet compared to previous oriented object detectors.

Our contributions in this article can be concluded in the following four points.

1) We propose FPDDet, which can effectively overcome the existing problems of exchangeability of edges, disorder of points, and complex regression parameters of polar detectors.
2) We put forward an SDR method based on a covariance-adaptive rotated Gaussian heatmap, which can reduce the dependency between heatmap and regression parameters. Moreover, we suggest a dense regression heatmap loss function to discriminate densely arranged targets.
3) We design the SAR image feature enhancement (FE) module to relatively enhance the target features and weaken the complex background.
4) Our approach achieves the state-of-the-art 97.82% map on RSSDD and 91.77% map on rotated ship detection dataset (RSDD). Compared to the classical oriented

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

object detectors, the results verify the superiority of our detector and the effectiveness of the proposed strategies.

II. Related Works

Object detectors have evolved at a rapid pace in recent years, and a large number of novel ideas have emerged. In this section, we discuss these related works in terms of the classification of HBB- and OBB-based detectors.

A. Horizontal Object Detection

The prediction accuracy of HBB-based detectors on publicly available datasets is rapidly increasing, and they can be generally classified into one- and two-stage detectors based on the presence or absence of proposal boxes. The focal loss (FL) used by RetinaNet [27] is proposed to balance the positive and negative sample weights, aiming to alleviate the sample imbalance problem; ATSS [28], YOLOX [29], YOLOE [30], and so on introduce different dynamic sample allocation strategies to achieve reasonable allocation of positive samples at different scales. As an anchor-free method, FCOS [31] is designed to solve the problem of too many low-quality bounding boxes, which adds centerness branch to reduce the weight of low-quality bounding boxes. CenterNet [32] and CornerNet [33] utilize heatmaps to predict the probability of each point in the feature map of the target. To address the issue of imbalanced sample allocation, Cascade RCNN [34] employs incremental IOU thresholds to interconnect multiple heads and achieve phased training.

For SAR image ship detection tasks, Sun et al. [35] proposed a category-position module and redesigned target classification and bounding box regression methods based on FCOS to improve the SAR ship object localization performance. Ma et al. [15] improved multiscale SAR ship detection by jumping and aggregating nodes and introduced channel attention to improve feature extraction. Zha et al. [16] proposed a locally enhanced network and feature fusion module to enhance the detection performance of small ship targets in SAR images, as well as optimized backbone networks to reduce the effect of background on detection.

The HBB-based ship detection methods described above produce highly overlapping prediction boxes, which causes difficulties in detecting densely arranged ships in inshore areas. To overcome this problem, the OBB methods have been developed.

B. Oriented Object Detection

The introduction of OBB-based detectors reduces the overlap between prediction boxes and effectively improves the ability of neural networks to predict densely aligned objects. Current methods of representing OBB include the mainstream five-parameter ($x, y, w, h,$ and α) representation and the eight-parameter representation of arbitrary quadrilateral vertices, the emerging polar coordinate representation, and so on.

To achieve oriented object detection, detectors improve the network structure based on the above OBB representation. IENet [24] combines direction prediction tasks with classification and regression using interactive modules. PIOU [21] calculates the IOU of the number of pixel points between the ground truth box and the predicted box as the loss of the sample and makes it derivable. Oriented RepPoints [36] suggest a representation of rotating targets using adaptive points and propose a quality evaluation and sample assignment scheme for adaptive point learning. To generate high-quality anchors, S2aNet [37] designed the feature alignment module and adaptively aligns features. The orientation detection module encodes orientation information to generate orientation-sensitive and orientation-invariant features. PolarDet [25] presents using the center, four polar angles, the ratio of the four polar diameters to the short side of the minimum enclosing box, and the short side to represent the rotation object. ROI Transformer [18] takes the learned rotational proposal boxes through RRoI warping to obtain rotationally invariant features. Oriented R-CNN [38] estimates the HBB and the offset of the two vertices of the OBB relative to the midpoint of the HBB in the RPN phase. ReDet [39] proposes to group convolutional kernels in channel dimensions and designs rotation-invariant ROIAlign to obtain rotation-equivalent features. To improve the classification score, OSKDet [40] proposes to represent OBB with a disordered orientation-sensitive heatmap and fuse key point localization distribution features.

For OBB-based SAR image ship detection tasks, the above models are widely applied in SAR oriented object detection, where oriented R-CNN can be easily transferred to SAR image detection and show excellent detection performance. In addition, Zhao et al. [41] designed a coarse-to-fine head structure to improve the detection accuracy of oriented SAR ship targets while constructing feature fusion and attention structures to enhance the features of multiscale targets in SAR images. Li and Lang [42] introduced a mixed convolutional channel attention mechanism to enhance target features in oriented SAR ship detection. To deal with the boundary discontinuity of the Cartesian representation, Gao et al. [43] utilized a set of vectors from the target centroid to the elliptical boundary of the SAR ship to represent the bounding box. To solve the boundary problem of mainstream OBB methods, Li et al. [44] used a centroid and a set of vectors from the centroid to the object boundary to represent SAR ship targets as a way.

The above methods show excellent performance in predicting densely aligned targets in arbitrary directions. However, these detectors typically suffer from five- and eight-parameter boundary discontinuities or require more complex OBB representations to address the issue.

III. Proposed Method

In this section, we will present our proposed approaches as follows. First, we will show the overall network structure of FPDDet in Section III-A. Second, the FE network will be described in Section III-B. Then, we will introduce our five-parameter polar coordinate encoding and decoding algorithms in Section III-C. In Section III-D, we will introduce the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

dense regression strategy in detail. Finally, Section III-E will describe the design of the loss functions.

A. Pipeline

Our approach to representing the OBB maintains the advantages of the polar method, while the OBB can be represented with fewer parameters and in a more convenient way. The proposed method differs from CenterNet, which uses a common Gaussian heatmap to represent the target confidence, and we use an oriented Gaussian heatmap to further reduce the background interference. Unlike PolarDet and polar encodings, which use single-point regression, we adopt a dense regression strategy that relies on a confidence map to reduce the dominance of one point on the prediction result. We also introduce an attention mechanism to weaken the background noise while enhancing the target features. As shown in Fig. 3, the overall structure of FPDDet is divided into three parts, which will be described in detail as follows.

The backbone part of the neural network adopts ResNet50 and takes its top three feature layers $\{F_3, F_4, F_5\}$ as outputs. Considering the large variation of ship object scales in SAR images, we performed feature fusion on the output feature layers of different sizes from the backbone network to better extract the ship features of different scales. The specific operation is to upsample the feature maps using bilinear interpolation, then adjust the channels by 3×3 convolution (conv) and concatenate the feature maps of the next layer, and finally output the fused features by adjusting the number of channels by 3×3 conv. The feature fusion process is given as follows:

$$F'_{i-1} = \mathrm{FF}(F_i, F_{i-1}) \tag{1}$$

where FF denotes feature fusion in two feature maps of different sizes as follows:

$$\begin{aligned}&\mathrm{FF}(F_i, F_{i-1})\\ &\quad = \mathrm{Conv}_{3\times3}(\mathrm{Concat}(\mathrm{Conv}_{3\times3}(\mathrm{Bilinear}(F_i)), F_{i-1})).\end{aligned} \tag{2}$$

For the output after feature fusion, we have

$$F'_3 = \mathrm{FF}(\mathrm{FF}(F_5, F_4), F_3). \tag{3}$$

In the FE module, unlike previous methods [25], [26], [43] using CenterNet as the framework, which take the feature map with one-fourth the resolution of the input image as their input, we designate the feature map obtained through downsampling the input image by a factor of one-eighth as the output, which will effectively speed up the inference of the CNN without sacrificing prediction accuracy. We then use the feature maps that have passed through the spatial and channel attention architecture as input to the heads and perform a 1×1 conv before inputting the feature map to each head. Our three heads are consistent, with two 3×3 conv and one 1×1 conv regression heatmap, centroid offset, polar diameter, and polar angles. Heads output the heatmap with values from $H \in [0, 1]^{H' \times W' \times 1}$, offset $O \in \mathbb{R}^{H' \times W' \times 2}$, and polar $P \in \mathbb{R}^{H' \times W' \times 3}$. The head can be defined as follows:

$$\mathrm{Head} = \mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{3\times3}\left(F'_3\right)\right)\right). \tag{4}$$

B. Feature Enhancement Module

The presence of inherent speckle noise in SAR imagery and the complex background along the shore impede accurate prediction of the heatmap. To limit the above interferences while enhancing the position information of the ship target in the feature map, we introduce the position attention [44], [45] and channel attention [46] modules sequentially in the output of the feature fusion structure. The specific details of the attention module used in the model are shown in Fig. 3, where pixel-wise attention is applied to F'_i after the feature fusion module fuses objects of different scales. First, the features are adjusted by three 1×1 conv channels to obtain feature maps F'_{i1}, F'_{i2}, and F'_{i3}, where the channels of F'_{i1}, and F'_{i2} are reduced to one-eighth of the original, and a dot product operation is applied to obtain the feature matrix after reshaping. Then, the feature matrix is dot-produced after the softmax function with F'_{i3}, and finally, a dynamic parameter-weighted feature is added to the input element-wise. The process of input features can be represented by

$$\mathrm{Att}_{\mathrm{position}} = F'_i + W_d \times \left(F'_{i3} \cdot \mathrm{soft\,max}\left(F'^{T}_{i1} \cdot F'_{i2}\right)^T\right) \tag{5}$$

where W_d denotes a dynamic parameter.

The global average pooling (GAP) is applied to the output of the position attention in the channel attention as the weight of F'_i in the channel dimension, which is calculated as

$$F_{\mathrm{GAP}} = \frac{1}{H' \times W'} \sum_{i=0}^{H'-1} \sum_{j=0}^{W'-1} \mathrm{Att}_{\mathrm{position}}(i, j). \tag{6}$$

Then, full connect (FC) operations are used after the GAP to learn the correlation between channels, and the process is

$$\mathrm{Att}_{\mathrm{channel}} = F'_i + \mathrm{FC}(\mathrm{LN}(\mathrm{Relu}(\mathrm{FC}(F_{\mathrm{GAP}})))) \tag{7}$$

where the first FC layer reduces the number of channels to a quarter of the original number and restores the number of channels in the following FC layer. The weights of the channels are obtained and then summed with the input F'_i at the element-wise after broadcast, and finally, a 1×1 conv operation is used as input to the prediction head.

C. Polar Encoding and Decoding

The OpenCV method uses $(x, y, w, h,$ and $\alpha)$ to represent the OBB, where α refers to the acute angle between the positive half of the x-axis and the target, whose range is $[0, 90°)$. The problem of periodicity of angle (POA) and exchange of edges (EOE) occurred by this method is shown in Fig. 4(a), both of which lead to sudden changes in loss even when the ground truth and predicted boxes have a large intersection over union (IOU). The POA problem also exists in the long side definition method, as shown in Fig. 4(b), which takes the long side as the width (w), the other side as height (h), and the angle (α) defined as the one with the smaller absolute value of the angle between the two sides, in the range $(-90°, 90°]$. The eight-parameter method, as shown in Fig. 4(c), faces the problem of loss abruptness due to corner point disorder. Fig. 4(d) shows that targets with large aspect ratios are sensitive to angle, and even small angular

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

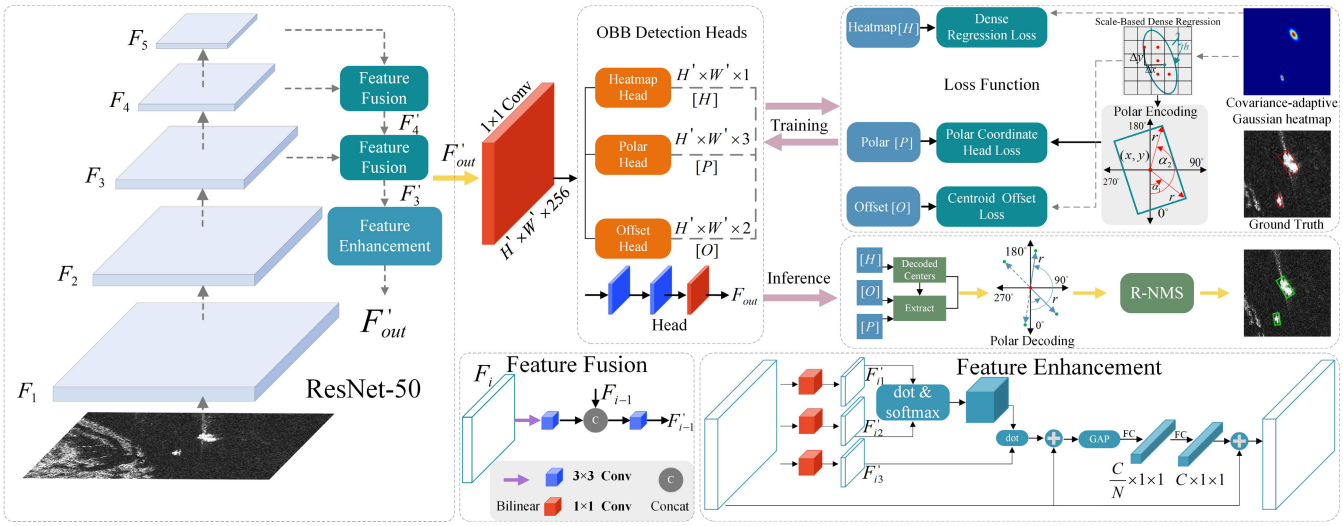


Fig. 3. Architecture of the proposed FPDDet. FPDDet consists of the backbone network, feature fusion structure, feature enhancement module, and output, which contains the heatmap head H, polar head P, and centroid offset head O. In the training phase, a scale-based dense regression strategy is introduced to train the network, using the generated covariance-adaptive rotated Gaussian heatmap to calculate the dense regression loss, P and O with the mapped polar coordinates and centroid offsets to calculate the polar coordinate head and centroid offset loss. In the inference phase, rotated nonmaximum suppression (R-NMS) is used to filter out overlapping prediction results.

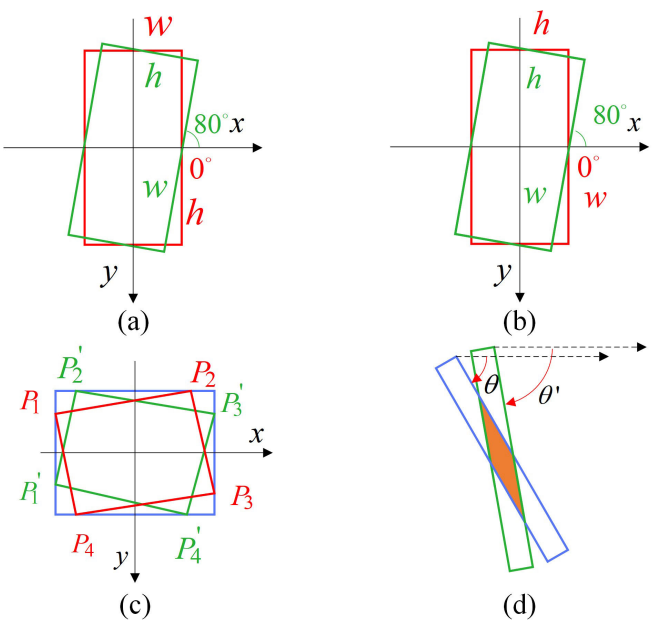


Fig. 4. Boundary problems of OBB representation methods in Cartesian coordinate system, where green boxes indicate prediction boxes and red indicates ground truth. (a) Periodicity of angle (POA) and exchange of edges (EOEs) problem of OpenCV representation method. (b) POA of the long-edge definition method. (c) Corner point disorder. (d) Small angular misalignment results in incorrect prediction for large aspect ratio target.

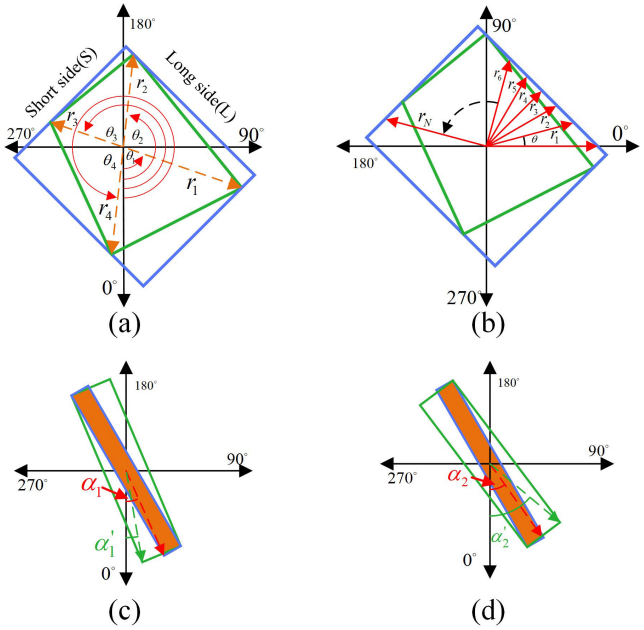


Fig. 5. (a) Polar representation of the target in PolarDet. (b) Polar representation of the target in polar encodings. (c) Effect of the offset of α_1 on IOU in our method. (d) Effect of the offset of α_2 on IOU in our method. In (a) and (b), the green boxes indicate the ground truth and the blue boxes indicate the minimum rectangular enclosing boxes, and in (c) and (d), the blue boxes indicate the ground truth and the green boxes indicate the prediction boxes.

misalignment can occur with large angle errors. All these issues present a challenge for network optimization.

The issues associated with representing oriented bounding boxes (OBBs) in Cartesian coordinates are largely resolved by using the polar coordinate system. As described in Fig. 5(a), PolarDet uses the center, four polar angles, ratios of four polar diameters to the short side of the smallest bounding box, and the short side to represent rotating objects. Polar encodings present a representation method using the centroid and a set of vectors from the centroid to the target boundary, as shown in Fig. 5(b). The above methods are able to solve the problems of edge exchangeability and corner point disorder well, while the polar coordinate system, as shown in Fig. 5(c) and (d), is better suited than the Cartesian coordinate system to cope with the sharp decrease in IOU for large aspect ratio targets due to angular offset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

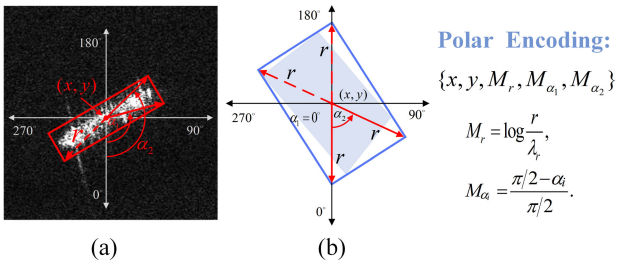


Fig. 6. Definition of five-parameter polar coordinate encoding. (a) Five-parameter polar coordinate representation of a ship target in arbitrary direction. (b) Encoding of the ship target with a zero polar angle. λ_r denotes the mean value of the short side of the minimum rectangular bounding box.

Considering the significant advantages of representing the OBB in the polar coordinate system, and for a more efficient and convenient representation of the OBB, we propose the encoding scheme shown in Fig. 6 as the encoding method for the ship target OBB. The point (x, y) represents the center of the smallest rectangular bounding box enclosing the target labels, the polar diameter r is the distance from the point (x, y) to the corner point of the rectangular box, and the polar angles (α_1 and α_2) are the angle between the corner point of the rectangular box and the polar axis in the range $[0°, 180°]$. The encoding of the regression parameters $(x, y, r, \alpha_1,$ and $\alpha_2)$ is listed in Algorithm 1.

In Algorithm 1, after obtaining the minimum rectangular bounding box $(x, y, w, h,$ and $\alpha)$ of the polygon label $L = \{L_i | L_i = (x_i, y_i), i = 1, 2, 3, 4\}$, the center of the rectangular box is further translated to the origin and the coordinates of the corner points $L' = \{L'_i | L'_i = (x'_i, y'_i), i = 1, 2, 3, 4\}$, are obtained. Suppose that the offset matrix of the corner points in the HBB with respect to the origin is $\sum = \begin{bmatrix} -w/2 & -w/2 & w/2 & w/2 \\ -h/2 & h/2 & -h/2 & h/2 \end{bmatrix}$, and the rotation matrix about the origin in Cartesian coordinates is $R = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, where the range of α is $(-90°, 90°]$. The corner point coordinate matrix calculated after the translation is

$$L' = \sum{}^{T} R. \tag{8}$$

Next, replace the Cartesian coordinate system representation of OBB with the polar diameter r and polar angles (α_1, α_2), where $r = (x_1'^2 + y_1'^2)^{1/2}$. As shown in Algorithm 1, multiple judgments are used to obtain polar angles to account for all possible cases of OBB. Considering that the case of a rectangular box with two corners on the vertical axis leads to the existence of three polar angles, we choose two small angles as polar angles.

In the regression phase of the parameters, based on the fact that our model is anchor-free frame, this leads us to directly regress the size of the polar diameter and not the offset between the polar diameter and the anchor box polar diameter, and the large-scale variation range of the polar diameter r poses a challenge to CNN stability, so we use the M_r function to map the polar diameter to play a role in reducing the range of polar diameter variation to stabilize the network, which can be calculated by

$$M_r = \log \frac{r}{\lambda_r} \tag{9}$$

Algorithm 1 Five-Parameter Polar Encoding

input : Polygon corner points of an OBB: $\{(x_i, y_i), i = 1, 2, 3, 4\}$, short edge mean $\boldsymbol{\lambda_r}$, offset limit value $\boldsymbol{\lambda_o}$

output: Regression values $\boldsymbol{f}$ after mapping five polar parameters $(x, y, r, \alpha_1, \alpha_2)$: $\boldsymbol{f} \leftarrow (M_r, M_{\alpha_1}, M_{\alpha_2}, M_o)$

Calculate the minimum rectangular bounding box (x, y, w, h, α) around corner points;

Calculate corner coordinates of rectangular box centered on origin $L' : \{(x'_i, y'_i), i = 1, 2, 3, 4\}$;

$L' \leftarrow \begin{bmatrix} \frac{-w}{2} & \frac{-w}{2} & \frac{w}{2} & \frac{w}{2} \\ \frac{-h}{2} & \frac{h}{2} & \frac{-h}{2} & \frac{h}{2} \end{bmatrix}^T \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$;

Calculate the polar diameter $r : r \leftarrow \sqrt{x_1'^2 + y_1'^2}$;

for $\langle x'_i, y'_i \rangle \in \{(x'_1, y'_1) \ldots, (x'_4, y'_4)\}$ **do**

  if $x'_i \geq 0$ & $y'_i > 0$ **then**

    $\beta_i \leftarrow \arctan \frac{x'_i}{y'_i}$;

  else if $x'_i > 0$ & $y'_i \leq 0$ **then**

    $\beta_i \leftarrow \frac{\pi}{2} + \arctan \frac{-y'_i}{x'_i}$;

  else if $x'_i = 0$ & $y'_i < 0$ **then**

    $\beta_i \leftarrow \pi$;

  end

end

$\beta \leftarrow (\beta_1, \beta_2, \ldots)$; Sort β in ascending order ; $\alpha_1 \leftarrow \beta_1$; $\alpha_2 \leftarrow \beta_2$;

Calculate the mapping M_r of the polar diameter $r : M_r \leftarrow \log \frac{r}{\lambda_r}$;

Calculate the mapping M_{α_i} for the polar angle $\alpha_i : M_{\alpha_i} \leftarrow \frac{\frac{\pi}{2} - \alpha_i}{\frac{\pi}{2}}$;

Calculate the centroid offset mapping M_o between the annotation center $'_{ann}$ and selected positive samples location $_{pos}$: $M_o \leftarrow \frac{center'_{ann} - location_{pos}}{\boldsymbol{\lambda_o}}$

Return $\boldsymbol{f} \leftarrow (M_r, M_{\alpha_1}, M_{\alpha_2}, M_o)$.

where λ_r in this method denotes the mean value of the short side h of the rectangular bounding box corresponding to the samples used during the whole CNN training process. The purpose of the logarithmic operation after dividing the polar diameter by the mean value λ_r includes the following:

1) further reducing the range of polar diameter variation for large targets;
2) mapping the polar diameter of small targets in the negative half-axis range while expanding the mapping range for small targets.

To solve the problem of normalizing the polar angle α_i of the same head regression as the polar diameter, we used the M_{α_i} function to simply map it to a space in the range $[-1, 1]$, which is calculated as follows:

$$M_{\alpha_i} = \frac{\pi/2 - \alpha_i}{\pi/2}. \tag{10}$$

Since we adopt a dense regression strategy, the values of polar diameter and polar angles assigned to the selected positive samples are consistent, but the values assigned to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

offset of the positive samples from the centroid of the ground truth are not consistent, we regress the offset M_o of the space locations $\text{location}_{\text{pos}}$ of each positive sample on the feature map from the centroid of the annotation after downsampling. For M_o, we have

$$M_o = \frac{\text{Center}'_{\text{ann}} - \text{location}_{\text{pos}}}{\lambda_o} \tag{11}$$

where $\text{Center}'_{\text{ann}}$ indicates that the annotation is downsampled to the centroid of the feature map space and λ_o is used to limit the range of offset variation (λ_o is set to 2).

The decoding process is shown in Algorithm 2, which decodes the predicted results of heatmap H, offset O, and polar P into the corner points of OBB. After obtaining the decoded OBB, the rotated nonmaximum suppression (R-NMS) operation was performed to remove the lower scoring boxes from the dense regression and finally output the prediction results of FPDDet.

Algorithm 2 Five-Parameter Polar Decoding

input : Predicted heatmap H, predicted center offset O, predicted polar diameter and polar angles P, downsampling ratio $\boldsymbol{r_s}$, short edge mean $\boldsymbol{\lambda}_r$, offset limit value $\boldsymbol{\lambda}_o$, heatmap score threshold $\boldsymbol{th}$.

output: Rectangular bounding box corner points: $\{(\hat{x}_i, \hat{y}_i), i = 1, 2, 3, 4\}$.

$(P_x, P_y) \leftarrow$ Find the coordinates of the points in H where H is greater than or equal to $\boldsymbol{th}$;

$(P_r, P_{\alpha 1}, P_{\alpha 2}, P_{xo}, P_{yo}) \leftarrow$ Find values in O, P that correspond to position (P_x, P_y);

Calculate decode center points (P'_x, P'_y), polar diameter P'_r, polar angles $(P'_{\alpha 1}, P'_{\alpha 2})$;

$P'_x \leftarrow \frac{P_x + \boldsymbol{\lambda}_o P_{xo}}{r_s}$; $P'_y \leftarrow \frac{P_y + \boldsymbol{\lambda}_o P_{yo}}{r_s}$;

$P'_r \leftarrow \boldsymbol{\lambda}_r \exp(P_r)$; $P'_{\alpha 1} \leftarrow \frac{\pi}{2} \times P_{\alpha 1}$; $P'_{\alpha 2} \leftarrow \frac{\pi}{2} \times P_{\alpha 2}$;

Calculate the coordinates of the polar points with polar diameter P'_r and polar angles $P'_{\alpha 1}$, $P'_{\alpha 2}$ in the Cartesian coordinate system;

$x_{\hat{1},2} \leftarrow P'_x + P'_r \cos(P'_{\alpha 1,2})$; $y_{1,2} \leftarrow P'_y + P'_r \sin(P'_{\alpha 1,2})$;

Return $\{(\hat{x_1}, \hat{y_1}), (\hat{x_2}, \hat{y_2}), (\hat{x_3}, \hat{y_3}), (\hat{x_4}, \hat{y_4})\}$; where the other two points are obtained by means of symmetry.

D. Scale-Based Dense Regression Strategy

The Centernet-based approach typically uses a key point as the positive sample of the target for the detector regression, which leads to an extremely strong correlation between the predicted heatmap and the polar coordinate feature map and the centroid offset feature map, even if there is a small deviation in their spatial location on the feature map, ultimately resulting in a false prediction of the target. Inspired by the idea of O^2-DNet [47] using dense regression, we propose a dense regression strategy based on a covariance-adaptive rotated Gaussian heatmap to dynamically assign the number of positive samples to ship objects. Unlike O^2-DNet, which uses all points within a fixed radius as positive samples, we assume that objects at different scales have different levels of attention.

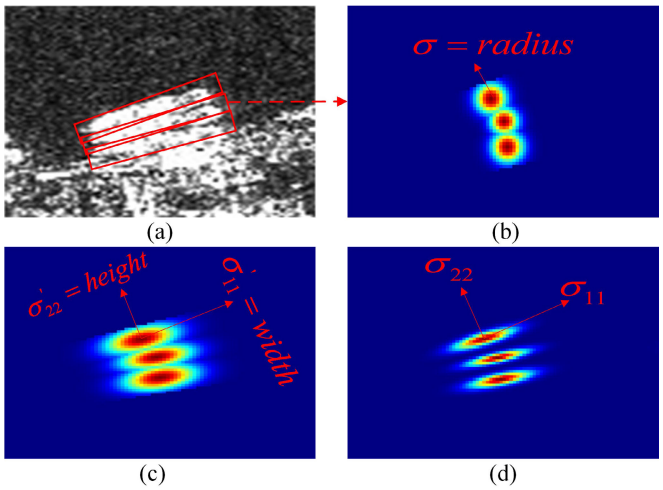


Fig. 7. (a) Input SAR ship image. (b) 1-D Gaussian heatmap with variance σ as the resulting Gaussian radius. (c) Rotated Gaussian heatmap where the covariance is the downsampled width and height of the ship. (d) Boundary decay to a fixed small value of the generated covariance-adaptive rotated Gaussian heatmap.

The implementation is given as follows: first, to generate the rotating Gaussian heatmap after downsampling the ship target, we have

$$M_h = R^T \begin{bmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{bmatrix} R \tag{12}$$

where R is the rotation matrix and σ_{11} and σ_{22} are the covariance of the 2-D Gaussian. The rotated Gaussian confidence map of the target is finally calculated by

$$\text{GM}(x, y) = \exp\left[-\frac{1}{2} \begin{pmatrix} x - x_c \\ y - y_c \end{pmatrix}^T M_h^{-1} \begin{pmatrix} x - x_c \\ y - y_c \end{pmatrix} \right] \tag{13}$$

where (x_c, y_c) is the object center.

Different heatmap generation strategies for densely arranged ship targets have a significant impact on the final prediction results. As shown in Fig. 7(b), CenterNet uses a 1-D Gaussian to generate the ship heatmap in arbitrary direction, which loses the spatial information of the ship; Fig. 7(c) uses the ship's width and height as the covariance of the rotated Gaussian heatmap, which retains the spatial information of the ship but does not significantly distinguish the ship's boundary, while introducing too much background may impede the decoding of the heatmap. Fig. 7(d) describes our proposed adaptive covariance-generated heatmap, retaining the spatial information of the ship targets while better distinguishing the ship boundaries.

In our experiment, adaptively generated covariances σ_{11} and σ_{22} are obtained by taking covariances corresponding to the confidence value of the heatmap when the boundary of ship target decays to a fixed small value λ_s (λ_s is set to 0.001), where the boundary is defined as the midpoints of the long and short sides of the rotated bounding box, namely, the adaptive covariances are covariances corresponding to the Gaussian heatmap when values at the midpoints of the short and long

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

sides of the target bounding box decay to λ_s. For boundary points P_b, we have

$$P_b = \begin{bmatrix} w/2 & 0 \\ 0 & h/2 \end{bmatrix} R. \tag{14}$$

Bringing the boundary points P_b into formula (13) gives

$$P_b M_h^{-1} P_b^T = \begin{bmatrix} -2\log^{\lambda_s} & 0 \\ 0 & -2\log^{\lambda_s} \end{bmatrix}. \tag{15}$$

Therefore, the covariance σ_{11} and σ_{22} can be calculated by (16)

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \end{bmatrix} = \begin{bmatrix} w^2/-8\log^{\lambda_s} \\ h^2/-8\log^{\lambda_s} \end{bmatrix}. \tag{16}$$

Then, we select the points whose heatmap values are greater than the threshold λ_{th} (λ_{th} is taken as 0.50) as the positive samples of the ships for regression. As shown in Fig. 8(b), unlike the classical CenterNet, which uses a single-point regression strategy, we assign multiple positive samples to the ship to reduce the strong correlation between the predicted heatmap and the polar coordinate feature map and the centroid offset feature map. Unlike O^2-DNet, as shown in Fig. 8(c) and (d), different amounts of positive samples are assigned to different scale targets, and the positive samples are optimized toward the center of the target.

E. Loss Function

The training loss contains three parts, corresponding to the rotated Gaussian heatmap head, the predicted polar diameter and polar angle head, and the relative object centroid offset head, where the loss of the heatmap uses a proposed dense regression-based heatmap loss (DRL) based on Gaussian FL (GFL) to cope with the single positive sample problem, compared to using cross entropy to learn the distance between the ground truth and the predicted target heatmap, we directly use smooth L1 loss [48] to learn the distance of the difference between the two, and the centroid offset head and the polar coordinate head use the smooth L1 loss to calculate the loss of the regression parameters.

For the rotated Gaussian heatmap branch, after the predicted values are obtained, the output is first normalized by the sigmoid operation, and then, the DRL between the heatmap H and the rotated Gaussian confidence map can be calculated as follows:

$$L_H = \frac{1}{N}\sum_{i=0}^{H'-1}\sum_{j=0}^{W'-1}\begin{cases} \left|P_{hij}^{\wedge} - \text{GM}_{ij}\right|^{\alpha^*} L_{\text{smooth}}(P_{hij}^{\wedge}, \text{GM}_{ij}) W_{hij}, & W_{hij} = 1 \\ -P_{hij}^{\wedge^{\alpha^*}} \log(1 - P_{hij}^{\wedge})(1 - W_{hij}), & \text{otherwise} \end{cases} \tag{17}$$

$$L_{\text{smooth}} = \begin{cases} \dfrac{0.5(P_{hij}^{\wedge} - \text{GM}_{ij})^2}{\beta^*}, & \left|P_{hij}^{\wedge} - \text{GM}_{ij}\right| < \beta^* \\ \left|P_{hij}^{\wedge} - \text{GM}_{ij}\right| - 0.5 \times \beta^*, & \text{otherwise} \end{cases} \tag{18}$$

where N denotes the number of positive samples selected, $P_{hij}^{\wedge} \in [0,1]^{H' \times W' \times 1}$ denotes the predicted confidence map, $\text{GM} \in [0,1]^{H' \times W' \times 1}$ denotes the encoded target rotated Gaussian heatmap, W_{hij} denotes the foreground weights (1 for foreground and 0 for background), and α^* and β^* are the two hyperparameters used for GFL and smooth L1 loss, which are set to 2 and 0.05, respectively, in the experiments.

For the centroid offset branch, smooth L1 loss is used to calculate the loss of the offset regression parameters and is implemented as follows:

$$L_O = \frac{1}{N}\sum_{i=1}^{N}\begin{cases} \dfrac{0.5\left(P_o^{\wedge} - M_o\right)^2}{\text{beta}}, & \left|P_o^{\wedge} - M_o\right| < \text{beta} \\ \left|P_o^{\wedge} - M_o\right| - 0.5 \times \text{beta}, & \text{otherwise} \end{cases} \tag{19}$$

where N is the number of positive samples, $P_o^{\wedge}$ is the predicted centroid offset, M_o is obtained from the difference between the location of the feature space where the positive samples are located and the target centroid location, and beta is set to 0.11 as a hyperparameter in the experiment.

For the polar coordinate head branch, the same smooth L1 loss is used to calculate the loss of polar diameter and polar angle, which is calculated by

$$L_P = \frac{1}{N}\sum_{i=1}^{N}\begin{cases} \dfrac{0.5(P_P^{\wedge} - M_P)^2}{\text{beta}}, & \left|P_P^{\wedge} - M_P\right| < \text{beta} \\ \left|P_P^{\wedge} - M_P\right| - 0.5 \times \text{beta}, & \text{otherwise} \end{cases} \tag{20}$$

where N is the number of positive samples, $P_P^{\wedge}$ is the predicted values of the polar head, M_P is obtained from the mapped polar diameter M_r and polar angles M_{α_i}, and beta is set to 0.11 in this loss function.

For the overall network loss, we have

$$L_{\text{all}} = L_H + L_O + L_P. \tag{21}$$

IV. Experiments and Analysis

A. Dataset and Implementation Details

To verify the effectiveness of FPDDet for SAR ship detection and its superiority over state-of-the-art detectors, we perform evaluations on the RSSDD [49] and RSDD [50] public datasets.

The RSSDD dataset comprises 1160 SAR images with 2540 ship targets and encompassing multiresolution images ranging from 271 × 241 to 526 × 646. In the experimental process, the dataset is divided into training and test sets with a ratio of 8:2; 928 images are used for training and 232 images are used for testing. To further evaluate the performance of the model, the test set is divided into 46 inshore images and 186 offshore images to separately test the detection capabilities of FPDDet under different SAR scenarios.

The RSDD dataset consists of 7000 slices of 512 × 512 resolution SAR images with 10 263 ship targets. In the experimental process, the dataset is divided into training and test sets with a ratio of 5:2; 5000 images containing 7271 ship objects are used for training and 2000 images containing 2992 ship objects are used for testing. In the testing phase,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

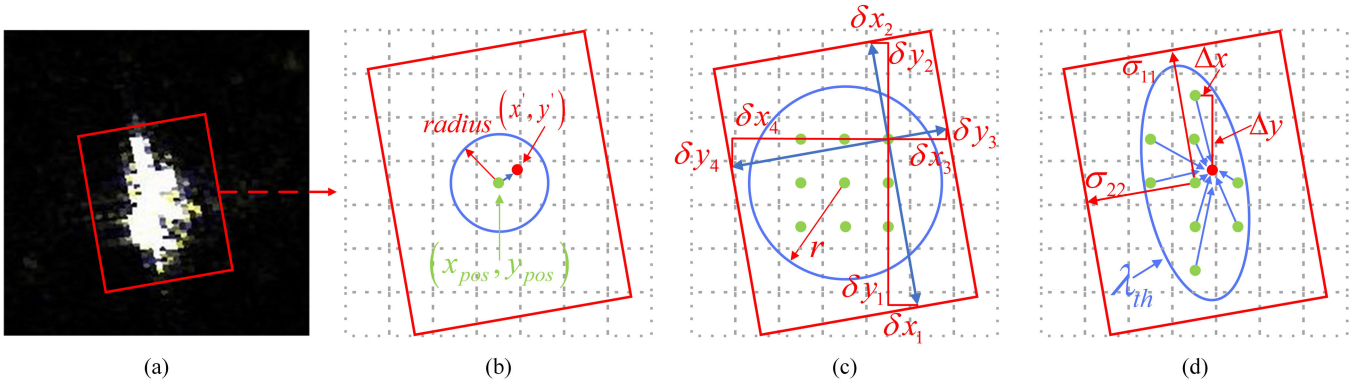


Fig. 8. Comparison of different regression strategies, where the green points indicate the feature map grid points and the red points indicate the target centroids after downsampling. (a) Input SAR ship image. (b) Single-point regression based on CenterNet. (c) Dense regression strategy of O^2-DNet, where r is a constant value and δ_{xi} and δ_{yi} denote the coordinate offsets from grid points to the target boundary. (d) Our proposed scale-based dense regression strategy, where λ_{th} is the heatmap score threshold for selecting positive samples, Δx and Δy are the horizontal and vertical coordinate offsets from grid points to the target centroid, respectively, and σ_{11} and σ_{22} are the covariances.

the test set is divided into 159 inshore images with 596 ship objects and 1841 offshore images with 2396 ship objects for detailed evaluation of the performances of the model in different scenarios.

In the training and testing phases of FPDDet, all experiments are implemented on an RTX 3060 GPU on the Pytorch platform. We trained 150 epochs using stochastic gradient descent (SGD) optimization with a batch size set to 2. The weight decay is set to 0.0005 and the momentum is set to 0.9. We use the cosine learning rate schedule with the initial learning rate set to 1.25×10^{-4}. During the training process, the RSSDD images are resized to a uniform 608×608 resolution and RSDD images maintain 512×512 resolution.

B. Ablation Study

We perform all ablation studies using the RSSDD dataset with ResNet-50 as the baseline, where precision–recall (PR) curve and AP metrics are used to evaluate the performance of the proposed methods, and by default, all experiments are conducted using the VOC2012 [51] AP.5 IOU metric as the benchmark. First, we conduct comparative experiments among different representation schemes to validate the superiority of our proposed five-parameter polar encoding. Second, we compare different regression strategies to demonstrate the validity of our SDR strategy. Moreover, we also evaluate the effectiveness of our proposed DRL function. Finally, an ablation study is conducted to showcase the efficacy of each design in FPDDet.

1) *Five-Parameter Polar Encoding:* In the regression branch of the polar diameter and polar angles, our FPDDet constrains the range of variation of the polar diameter with the short edge mean of the minimum bounding box of the ship target in the training set. As shown in Table I, we selected different schemes for comparative experiments to verify the effectiveness and superiority of the proposed strategy. Although direct regression of polar diameter improves AP by 1.85% compared to the mainstream method of long edge definition of OBB in the Cartesian coordinate system, the large-scale variation leads to instability of the network training. Using a larger polar diameter mean to limit the range of polar diameter variation can stabilize the training process and improve the AP by 3.75% while increasing the prediction difficulty of the network. For instance, in the case of large ships, such as coastal cruise ships, if the prediction value contains the same error, a higher limiting value will exacerbate the error and lead to a degradation of predictive accuracy. When using the long edge mean, the AP decreases by 0.74% compared to the polar mean scheme. Compared to the OBB method, our adopted short edge scheme shows excellent detection performance for ship targets in the RSSDD dataset, with a 10.09% AP improvement in the inshore and 3.15% AP improvement in the offshore, for an overall improvement of 5.16% AP, which proves that our five-parameter encoding in polar coordinate system and the polar diameter mapping scheme possess the significant advantages in rotating SAR ship localization.

2) *Scale-Based Dense Regression Strategy:* We compare the impact of different regression strategies on detectors, including CenterNet-based single-point regression, O^2-DNet-based positive-sample regression within a fixed radius (FRDR), dense regression with a rotated Gaussian heatmap generated by the width and height of the target as covariance (WHDR), and covariance-adaptive dense regression (CADR) strategies. As shown in Table I, compared to single-point regression, CADR improved the AP by 8.5% and 1.07% in the inshore and offshore scenarios, respectively, demonstrating that the prediction performance of the detector can be significantly improved by reducing the correlation between the predicted heatmap and the polar coordinate feature map and the centroid offset feature map. Compared to FRDR, the AP has been enhanced by 6.51% and 0.33% in the inshore and offshore scenarios, respectively, which shows the importance of a reasonable distribution of positive samples for targets of different scales. Compared to WHDR, the AP has been enhanced

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARATIVE RESULTS OF DIFFERENT REPRESENTATION SCHEMES IN DIFFERENT SCENARIOS

Backbone	scheme	Inshore			Offshore			All		
		Recall	Precision	AP.5	Recall	Precision	AP.5	Recall	Precision	AP.5
ResNet-50	Cartesian System	87.79	57.85	84.18	97.06	79.43	96.14	94.14	71.59	92.66
	Polar System	87.79	63.71	85.15	98.66	83.30	98.34	95.24	76.47	94.51
	Polar+Long Edge Mean	90.70	60.94	88.40	99.20	87.09	98.70	96.52	77.27	95.67
	Polar+Polar Diameter Mean	93.02	58.82	91.69	98.66	85.22	98.40	96.89	75.04	96.41
	Polar+Short Edge Mean	**95.35**	**86.32**	**94.27**	**99.47**	**95.38**	**99.29**	**98.17**	**92.41**	**97.82**

Bold items denote the optimal values in the columns.

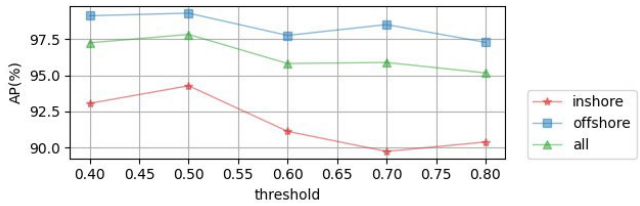


Fig. 9. Effect of different thresholds on the scale-based dense regression strategy.

TABLE II

COMPARISON OF DIFFERENT REGRESSION STRATEGIES

Method	Inshore			Offshore		
	Recall	Precision	AP.5	Recall	Precision	AP.5
Single Point	89.53	62.86	83.94	98.66	90.22	97.98
FRDR	88.95	59.07	85.93	99.20	89.61	98.71
WHDR	92.44	62.85	90.10	98.93	92.27	98.59
CADR (ours)	**93.02**	**80.81**	**92.44**	**99.20**	**96.61**	**99.05**

Bold items denote the optimal values in the columns.

by 2.34% and 0.46% in the inshore and offshore scenarios, respectively, verifying the effectiveness of the covariance-adaptive heatmap for the detection of densely arranged ship targets.

In SDR, we choose a threshold λ_{th} for the dynamic assignment of positive samples based on the score of the heatmap. To find the appropriate threshold, we use different values to do the comparative experiments. As shown in Fig. 9, when the threshold λ_{th} is set to 0.50, the AP is optimal in the inshore and offshore scenarios. When the threshold increases, the selected samples decrease and the dependence between the predicted heatmap, the polar coordinate feature map, and the centroid offset feature map increases, making the network sensitive to the target location. As the threshold decreases, more samples will be selected, which may introduce background noise into the bounding box.

3) *Dense Regression-Based Loss Function:* As described in Table III, the detector shows excellent detection performance when using GFL as the heatmap loss function for dense regression; however, as shown in Fig. 10, in dense regression, the positive samples are all optimized toward 1, which leads to the problem of not correctly taking out the local maximum to decode the regression parameters. To be able to optimize the samples towards different confidence values in dense regression, the DRL function directly learns the difference distance between the prediction and the target heatmap to cope with the above problem. Meanwhile, in Table III, we compare the detection performance of four loss functions for different SAR image scenes: FL, binary cross entropy (BCE), GFL, and DRL. Among them, DRL improves by 0.29% in the inshore scenes and 0.18% in the offshore scenes compared to GFL,

TABLE III

COMPARISON OF DIFFERENT HEATMAP LOSS FUNCTIONS

Method	Inshore			Offshore		
	Recall	Precision	AP.5	Recall	Precision	AP.5
FL [27]	91.86	53.2	89.67	98.13	88.01	97.35
BCE	84.88	61.09	81.17	97.59	90.12	97.09
GFL [33]	94.77	72.44	93.98	99.2	88.54	99.11
DRL (ours)	**95.35**	**86.32**	**94.27**	**99.47**	**95.38**	**99.29**

Bold items denote the optimal values in the columns.

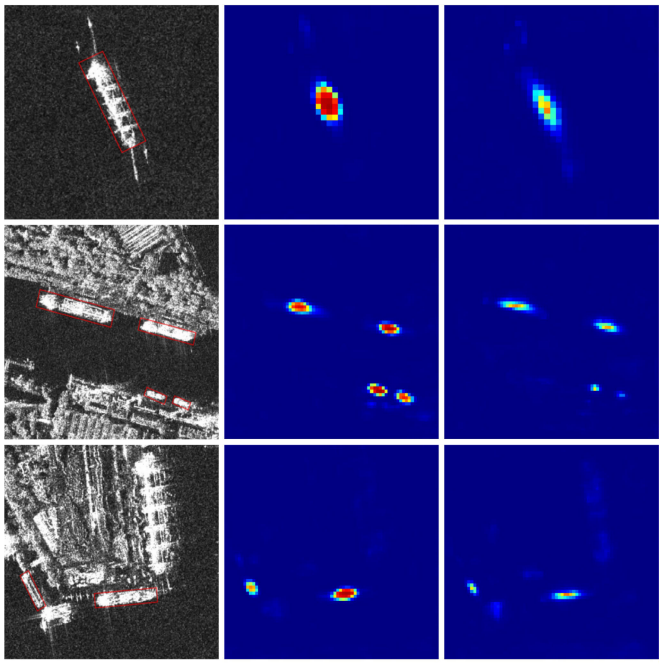


Fig. 10. (a) Input SAR images at the time of prediction, red boxes are ground truth. (b) Predicted ship heatmap using Gaussian focal loss. (c) Ship heatmap predicted by the proposed dense regression-based heatmap loss function.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION EXPERIMENT OF FPDDET ON RSSDD WITH DIFFERENT DESIGNS

Method	FP	SDR	FE	DRL	AP.5	FPS
FPDDet					92.66	15.2
	√				93.52	15.2
	√	√			96.13	14.8
	√		√		93.54	14.6
	√	√	√		97.57	14.6
	√	√	√	√	**97.82**	14.6

Bold items denote the optimal values in the columns.

indicating that optimizing positive samples toward different confidence values in dense regression can improve the detection performance.

4) *Ablation Experiments:* Table IV shows the ablation study of each design in the proposed FPDDet, including the comparison of five-parameter polar coordinate encoding (FR), SDR, FE module, and DRL function. Compared with the target representation in the Cartesian coordinate system, our five-parameter polar encoding achieves 0.86% AP gain while maintaining the same decoding speed. The introduction of the SDR strategy improves AP by 2.61% with a slight reduction in the prediction speed. In addition, the introduction of the FE module effectively improves the detection performance of ship targets in complex backgrounds, achieving 1.44% AP gain. As shown in Fig. 10, the improved dense regression heatmap loss function effectively addresses the positive sample optimization problem of GFL in performing dense regression heatmap prediction, and the predicted spatial feature of ship targets is more hierarchical and boundaries are clearer. Overall, ablation experiments have verified that each design in FPDDet is conducive to the SAR ship detection and our full model achieves the best balance across all the metrics.

C. Comparison With the State-of-the-Art Methods

To verify the effectiveness of our method, comparison results with several state-of-the-art object detectors are given, including anchor-based one-stage methods, reppoint-based anchor-free detectors, polar-based object representation, and two-stage detectors.

1) *Anchor-Based One-Stage Methods:* Rotated retinanet as a classical one-stage method, the proposed focal loss effectively addresses the problem of imbalance between positive and negative sample weights in the prediction of category scores in the network. S2aNet extracts rotation-sensitive features in the refinement stage and pools to produce rotation-invariant features to address the accurate prediction of objects in arbitrary orientations.
2) *Reppoint-Based Anchor-Free Methods:* Reppoint-based methods use convex hulls formed by point sets to predict object bounding boxes, which are not limited by the size of anchor boxes and show superior detection performance for objects of different shapes. Sasm reppoints [52] adaptively generate different IOUs for objects of different scales and evaluate different positive sample weights. Oriented reppoints select positive samples by evaluating adaptive points in terms of score, localization, orientation, and point-wise correlation.
3) *Polar-Based Object Representation:* Polar encodings represent the oriented object with a set of polar coordinate vectors, which effectively addresses the boundary problem of mainstream OBB-based detectors.
4) *Two-Stage Detectors:* ReDet extracts the rotational equivariant features of arbitrary direction targets by grouping channels, which effectively preserves the overall features of arbitrary direction targets. Oriented R-CNN proposes an efficient oriented RPN to generate high-quality oriented proposals.

Table V shows the test results in both inshore and offshore scenarios. When baselined with ResNet-50, FPDDet achieves a balance between accuracy and prediction speed in experiments that involve different ResNets as backbones. FPDDet achieves 97.82% AP in RSSDD test sets, which outperforms all previous state-of-the-art detectors and achieves an increase of 1.20% AP over the second highest performing detector, Oriented R-CNN. In particular, in the inshore test scenario, FPDDet achieves an AP gain of 6.70% relative to single-stage methods and 4.48% relative to two-stage detectors. Specifically, FPDDet achieves an AP gain of 16.49% in the inshore scenario, 0.70% gain in the offshore scenario, and 5.29% across all scenarios compared to polar encodings method while maintaining high detection speed.

To further verify the detection performance of FPDDet, Fig. 11 shows the PR curves of our method and other oriented object detectors in inshore and offshore scenarios. As shown in Fig. 11(a), it is obvious that the PR curve of FPDDet is superior to those of other state-of-the-art detectors in the inshore scenario, demonstrating the effectiveness and superiority of our proposed detector in dealing with situations, such as complex inshore backgrounds and dense arrangements of ship targets. For the offshore scenario, as shown in Fig. 11(b), Oriented R-CNN and our method show superior detection performance compared to other methods. As shown in Fig. 11(c), our PR curve is the highest compared to other state-of-the-art detectors in all test sets, further confirming the advantage of our method for SAR ship detection.

To qualitatively verify the superiority of FPDDet, Figs. 12 and 13 show the visual comparative results in the RSSDD dataset, which further demonstrates the anti-interference performance of FPDDet against complex backgrounds, superiority of detecting densely aligned ship targets, and accuracy of predicting small targets in offshore scenarios. Compared with the normal OBB encoding, the polar coordinate encoding makes the FPDDet more accurate in detecting ships with large aspect ratios and reduces both false positives and false negatives in SAR ship detection.

D. Experimental Results on RSDD Dataset

Table VI presents the detection results of FPDDet with detectors such as two-stage, one-stage, and anchor-free for different scenarios of RSDD. By using ResNet-101 as the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISON OF RESULTS OF DIFFERENT METHODS FOR INSHORE AND OFFSHORE SCENARIOS OF RSSDD

Method	Inshore			Offshore			All			FPS
	Recall	Precision	AP.5	Recall	Precision	AP.5	Recall	Precision	AP.5	
Rotated retinanet [27]	84.88	47.71	74.75	98.93	84.47	98.61	94.51	69.35	92.38	16.7
S2aNet [37]	89.53	41.73	87.23	98.93	86.85	98.77	95.97	65.91	95.46	15.2
Sasm repppoints [52]	80.23	38.55	73.75	99.20	87.50	99.04	93.22	65.09	92.15	15
Oriented reppoints [36]	89.53	49.52	87.57	98.93	92.04	98.75	95.97	73.49	95.53	15.3
Polar Encodings [26]	81.98	68.45	77.78	98.93	95.12	98.59	93.59	85.88	92.53	9.3
ReDet [39]	90.12	81.15	88.57	99.20	96.36	99.09	96.34	91.32	95.98	1.7
Oriented R-CNN [38]	90.70	75.36	89.79	**99.73**	**96.38**	**99.63**	96.89	89.66	96.62	13.8
FPDDet (ResNet-34)	89.53	86.03	87.98	98.13	94.10	97.11	95.42	91.56	94.29	**18.8**
FPDDet (ResNet-50)	**95.35**	**86.32**	**94.27**	99.47	95.38	99.29	**98.17**	**92.41**	**97.82**	14.6
FPDDet (ResNet-101)	93.60	83.85	92.07	99.47	92.54	99.34	97.62	89.73	97.20	11.4

Bold items denote the optimal values in the columns, the underlined items represent the second-best values in the columns.

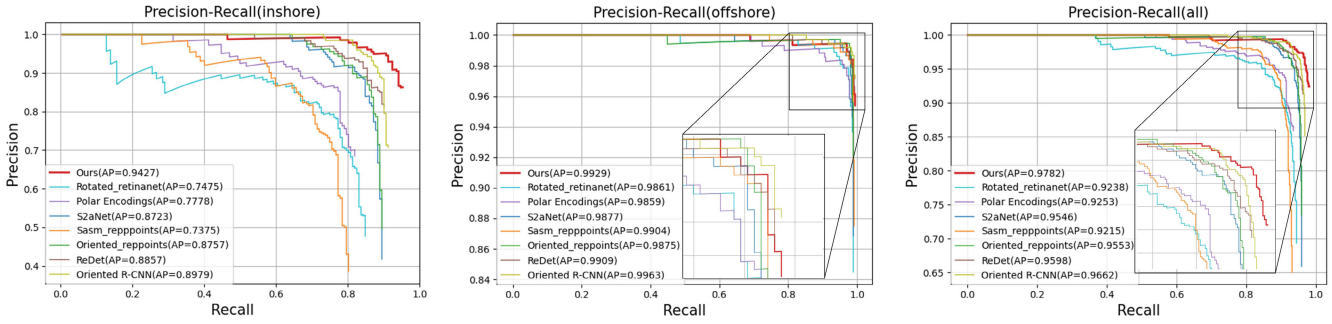


Fig. 11. PR curves of FPDDet and comparison methods. (a) PR curves of RSSDD inshore scene. (b) PR curves of RSSDD offshore scene. (c) PR curves of RSSDD all scenes.

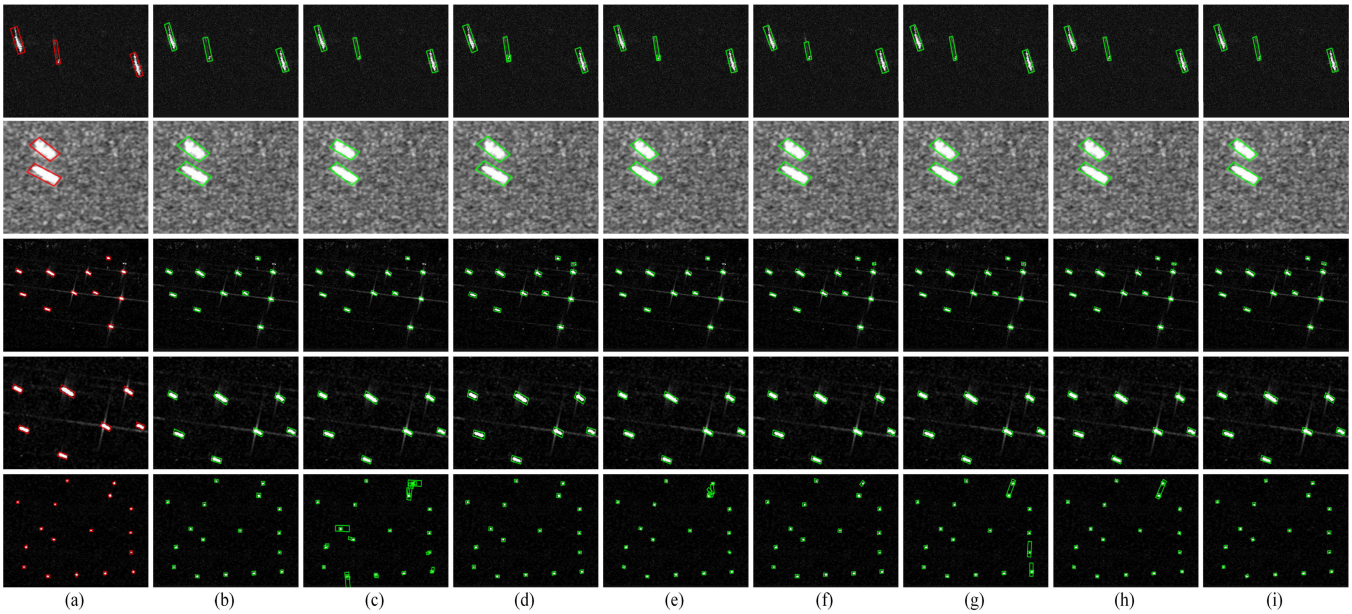


Fig. 12. Visualization results of different detectors in the RSSDD offshore scenario. (a) Ground truth. (b) Our method. (c) Rotated retinanet. (d) Polar encodings. (e) S2aNet. (f) Sasm repppoints. (g) Oriented reppoints. (h) ReDet. (i) Oriented R-CNN.

baseline, our model has achieved a 91.77% AP optimal outcome. Compared to the polar encodings model, we obtained a 14.65% AP improvement in the inshore scenario, 5.15% AP improvement in the offshore scenario, and 3.89% AP improvement in all scenarios. When compared to the previous best RSDD detector, S2aNet, our detector showed a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

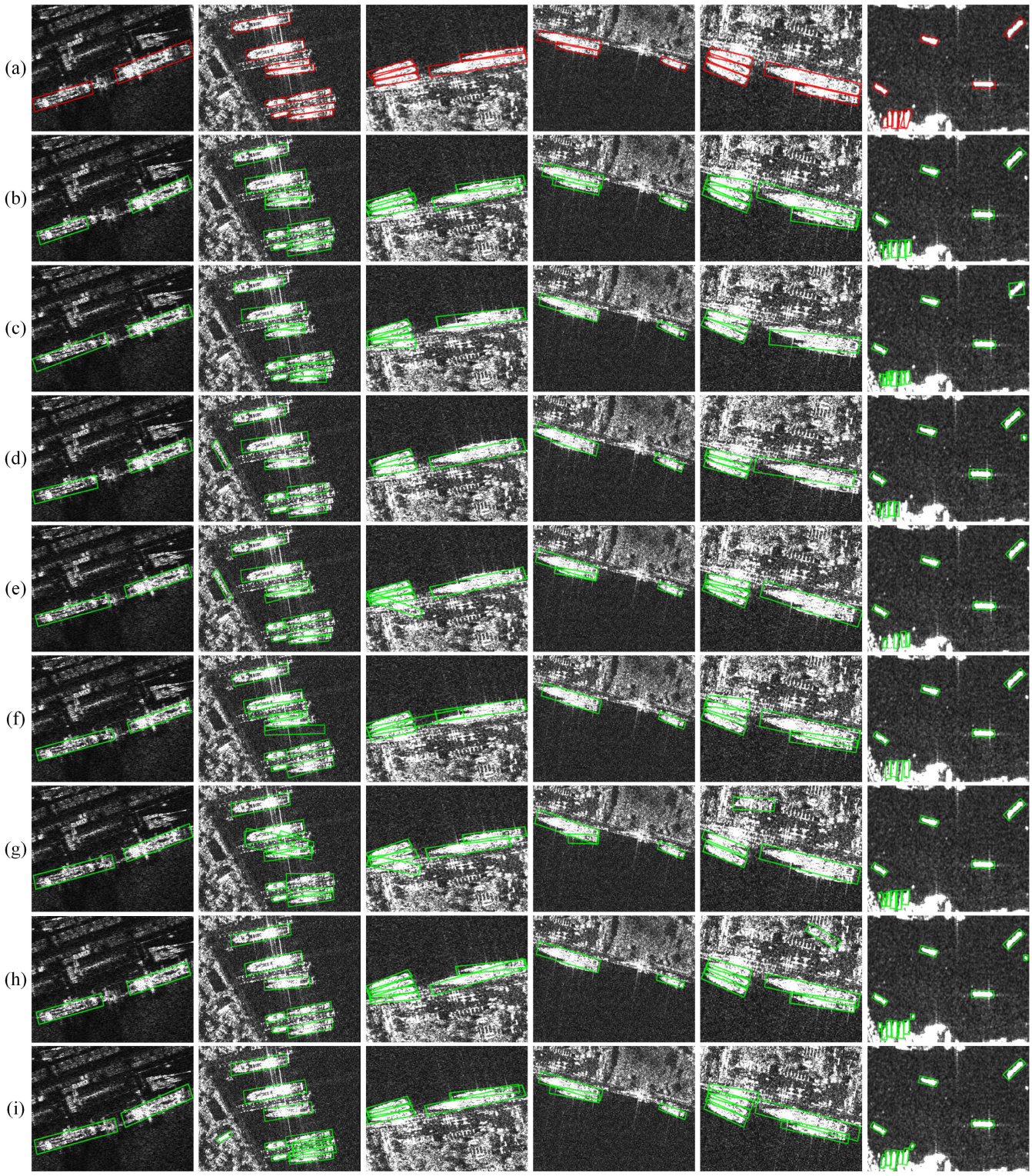


Fig. 13. Visualization results of different detectors in the RSSDD inshore scenario. (a) Ground truth. (b) Our method. (c) Rotated retinanet. (d) Polar encodings. (e) S2aNet. (f) Sasm repppoints. (g) Oriented reppoints. (h) ReDet. (i) Oriented R-CNN.

10.24% AP improvement in the inshore scenario, a 0.20% AP improvement in the offshore scenario, and a 1.71% improvement in all scenarios. The detection outcomes of the RSDD dataset indicate that FPDDet holds a significant advantage for ship detection in the inshore complex scenario while maintaining high detection accuracy in the offshore. This result is consistent with the RSSDD detection results. In addition, the RSDD detection outcomes confirm that our FPDDet can achieve superior performance in various SAR image environments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF RESULTS OF DIFFERENT METHODS FOR INSHORE AND OFFSHORE SCENARIOS OF RSDD

Method		Backbone	Inshore AP.5	Offshore AP.5	All AP.5
two-stage method	Oriented R-CNN [38]	ResNet-50	65.92	90.21	88.84
		ResNet-101	66.77	90.28	88.85
	ReDet [39]	ReResNet-50	61.94	90.34	88.40
one-stage method	Rotated retinanet [27]	ResNet-50	33.20	74.06	66.66
		ResNet-101	35.75	74.92	67.89
	S2aNet [37]	ResNet-50	63.27	93.14	87.91
		ResNet-101	66.43	<u>94.94</u>	90.06
	R3Det [19]	ResNet-50	56.87	90.16	80.87
		ResNet-101	57.73	90.09	80.92
anchor-free method	Polar Encodings [26]	ResNet-50	56.69	90.12	87.31
		ResNet-101	62.02	89.99	87.88
	FPDDet(ours)	ResNet-50	<u>76.08</u>	94.29	<u>91.05</u>
		ResNet-101	**76.67**	**95.14**	**91.77**

Bold items denote the optimal values in the columns, the underlined items represent the second-best values in the columns.

V. CONCLUSION

This article proposes an oriented SAR ship detector named FPDDet, which represents SAR ship targets in the polar coordinate system. Five polar coordinate parameters are used to represent ship targets conveniently and effectively to cope with the boundary discontinuity problem and the IOU drop problem of large aspect ratio objects that exist in the current mainstream OBB-based detectors. Meanwhile, to mitigate the correlation between the predicted heatmap, the polar coordinate feature map, and the centroid offset feature map, we propose an SDR strategy for dynamically assigning positive samples to ship targets of different scales. To improve the detection performance of the detector for complex inshore scenes, we put forward a covariance-adaptive heatmap to cope with the dense alignment of ship targets. In addition, we improve the heatmap loss function to learn the target foreground and introduce an attention mechanism to suppress the complex background. Experimental results on the RSSDD and RSDD datasets show that FPDDet achieves 97.82% AP and 91.77% AP, respectively, which is an increase of 1.20% AP and 1.71% over the second highest performing detector.

REFERENCES

[1] D. Cerutti-Maori, J. Klare, A. R. Brenner, and J. H. G. Ender, "Wide-area traffic monitoring with the SAR/GMTI system PAMIR," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 10, pp. 3019–3030, Oct. 2008.

[2] A. Renga, M. D. Graziano, and A. Moccia, "Segmentation of marine SAR images by sublook analysis and application to sea traffic monitoring," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1463–1477, Mar. 2019.

[3] W. Ao, F. Xu, Y. Li, and H. Wang, "Detection and discrimination of ship targets in complex background from spaceborne ALOS-2 SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 2, pp. 536–550, Feb. 2018.

[4] S. Schlaffer, P. Matgen, M. Hollaus, and W. Wagner, "Flood detection from multi-temporal SAR data using harmonic analysis and change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 38, pp. 15–24, Jun. 2015.

[5] X. Leng, K. Ji, X. Xing, S. Zhou, and H. Zou, "Area ratio invariant feature group for ship detection in SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2376–2388, Jul. 2018.

[6] G. Gao, K. Ouyang, Y. Luo, S. Liang, and S. Zhou, "Scheme of parameter estimation for generalized gamma distribution and its application to ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1812–1832, Mar. 2017.

[7] O. Pappas, A. Achim, and D. Bull, "Superpixel-level CFAR detectors for ship detection in SAR imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1397–1401, Sep. 2018.

[8] Z. Wang, R. Wang, X. Fu, and K. Xia, "Unsupervised ship detection for single-channel SAR images based on multiscale saliency and complex signal kurtosis," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4011305.

[9] S. Song and J. Yang, "Ship detection in polarimetric SAR images via tensor robust principle component analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 3152–3155.

[10] C. Wang, J. Shi, Z. Zou, W. Wang, Y. Zhou, and X. Yang, "A semi-supervised SAR ship detection framework via label propagation and consistent augmentation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 4884–4887.

[11] M. Chendeb El Rai, J. H. Giraldo, M. Al-Saad, M. Darweech, and T. Bouwmans, "SemiSegSAR: A semi-supervised segmentation algorithm for ship SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[12] B. Hou et al., "A neural network based on consistency learning and adversarial learning for semisupervised synthetic aperture radar ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5220816.

[13] S. Chen, R. Zhan, W. Wang, and J. Zhang, "Domain adaptation for semi-supervised ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] Y. Zhou, X. Jiang, Z. Chen, L. Chen, and X. Liu, "A semisupervised arbitrary-oriented SAR ship detection network based on interference consistency learning and pseudolabel calibration," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5893–5904, 2023.

[15] X. Ma, S. Hou, Y. Wang, J. Wang, and H. Wang, "Multiscale and dense ship detection in SAR images based on key-point estimation and attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5221111.

[16] M. Zha, W. Qian, W. Yang, and Y. Xu, "Multifeature transformation and fusion-based ship detection with small targets and complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[17] L. Bai, C. Yao, Z. Ye, D. Xue, X. Lin, and M. Hui, "Feature enhancement pyramid and shallow feature reconstruction network for SAR ship detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1042–1056, 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.

[18] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[19] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 4, pp. 3163–3171.

[20] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8232–8241.

[21] Z. Chen et al., "PIoU Loss: Towards accurate oriented object detection in complex environments," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 195–211.

[22] K. Fu, Z. Chang, Y. Zhang, and X. Sun, "Point-based estimator for arbitrary-oriented object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4370–4387, May 2021.

[23] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[24] Y. Lin, P. Feng, J. Guan, W. Wang, and J. Chambers, "IENet: Interacting embranchment one stage anchor free detector for orientation aerial object detection," 2019, *arXiv:1912.00969*.

[25] P. Zhao, Z. Qu, Y. Bu, W. Tan, and Q. Guan, "PolarDet: A fast, more precise detector for rotated target in aerial images," *Int. J. Remote Sens.*, vol. 42, no. 15, pp. 5831–5861, Aug. 2021.

[26] Y. He, F. Gao, J. Wang, A. Hussain, E. Yang, and H. Zhou, "Learning polar encodings for arbitrary-oriented ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3846–3859, 2021.

[27] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[28] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[29] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[30] S. Xu et al., "PP-YOLOE: An evolved version of YOLO," 2022, *arXiv:2203.16250*.

[31] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[32] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[33] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 734–750.

[34] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[35] Z. Sun et al., "An anchor-free detection method for ship targets in high-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7799–7816, 2021.

[36] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented RepPoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1829–1838.

[37] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[38] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[39] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[40] D. Lu, D. Li, Y. Li, and S. Wang, "OSKDet: Orientation-sensitive keypoint localization for rotated object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1182–1192.

[41] S. Zhao, Q. Liu, W. Yu, and J. Lv, "A single-stage arbitrary-oriented detector based on multiscale feature fusion and calibration for SAR ship detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8179–8198, 2022.

[42] C. Li and H. Lang, "An oriented SAR ship detector with mixed convolution channel attention module and geometric nonmaximum suppression," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8074–8084, 2022.

[43] F. Gao, Y. Huo, J. Sun, T. Yu, A. Hussain, and H. Zhou, "Ellipse encoding for arbitrary-oriented SAR ship detection based on dynamic key points," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5240528.

[44] C. Li, C. Xu, Z. Cui, D. Wang, T. Zhang, and J. Yang, "Feature-attentioned object detection in remote sensing imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 3886–3890.

[45] Y. Li, Q. Huang, X. Pei, L. Jiao, and R. Shang, "RADet: Refine feature pyramid network and multi-layer attention network for arbitrary-oriented object detection of remote sensing images," *Remote Sens.*, vol. 12, no. 3, p. 389, Sep. 2020.

[46] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[47] H. Wei, Y. Zhang, Z. Chang, H. Li, H. Wang, and X. Sun, "Oriented objects as pairs of middle lines," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 268–279, Nov. 2020.

[48] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[49] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[50] C. Xu et al., "RSDD-SAR: Rotated ship detection dataset in SAR images," *J. Radars*, vol. 11, no. 4, pp. 581–599, 2022.

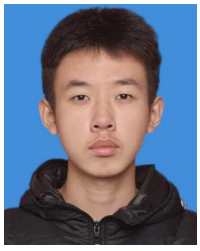
[51] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.

[52] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 923–932.



Moran Ju received the B.Eng. and M.Eng. degrees from Jilin University, Changchun, China, and the Ph.D. degree from the Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China, in 2012 and 2015, respectively.

He is currently a Lecturer with the College of Information Science and Technology, Dalian Maritime University, Dalian, China. His research interests include target detection, target tracking, and image processing.



Buniu Niu received the B.E. degree in electronic information engineering from Xi'an Technological University, Xi'an, China, in 2022. He is currently pursuing the master's degree with Dalian Maritime University, Dalian, China.

His research interests include deep learning and rotating object detection in synthetic aperture radar (SAR) images.



Jingbo Zhang received the B.Eng. and Ph.D. degrees from Dalian Maritime University, Dalian, China, in 2004 and 2012, respectively.

He is currently an Associate Professor with the College of Information Science and Technology, Dalian Maritime University, where he is mainly engaged in research on intelligent wireless perception. His research interests include physical layer security, cognitive radio, and blind modulation recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:44:50 UTC from IEEE Xplore. Restrictions apply.